

ANDY YANG

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TECHNICAL SKILLS

Languages: C++, Python, Go, TypeScript, JavaScript, SQL

Frameworks/Libraries: PyTorch, React, React Redux, Express.js, OpenCV, MediaPipe

Developer Tools/Databases: Git, Docker, Linux, Figma, MCP, Playwright, Vitest, MongoDB, ClickHouse, MySQL

EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

September 2022 - April 2028 (Expected)

EXPERIENCE

Software Engineer

Rundoo - AI Team (Internship)

September 2026 - Present

Redwood City, CA

- Building **agent evaluation frameworks** for Rundoo's proprietary AI agent to validate autonomous task execution

Software Engineer

BitGo - Web Experience Team (Internship)

May 2026 - August 2026

Toronto, ON

- Engineered an **agentic design-to-code pipeline** utilizing **Figma Code Connect**, **MCP**, and custom **agent skills**, enabling designers to **autonomously implement production-ready components** and fully own the design system
- Authored **500+ Playwright/Vitest visual tests** and reconciled UI components to structurally and visually match Figma
- Configured **Figma Code Connect** and set up mappings to structurally enforce design-to-code sync and **prevent drift**
- Containerized VRT baseline generation via **Docker** to resolve macOS/Linux rendering diffs, unlocking **Linux** self-hosted CI runners to reduce job latency by **>75%**

Backend Engineer

Tencent - WeChat Search Team (Internship)

April 2025 - November 2025

Beijing, China

- Designed and implemented a **Last Finishing Child based critical path algorithm** in **C++** to pinpoint bottlenecks in **RPC** traces, enabling **targeted debugging and optimization**
- Engineered a **pipelined trace-aggregation system** to ingest millions of daily traces in **chunked streams**, constructing dependency graphs to calculate critical paths within **strict container memory limits**
- Implemented **log correlation logic** in the logging pipeline to collect logs by request ID, match caller-callee RPC log pairs, and calculate and report request-level metrics, enabling **real-time performance monitoring per module**

Software Engineer

SGCC Big Data Center (Internship)

January 2025 - April 2025

Beijing, China

- Engineered a **RAG chat pipeline** in **Python** with **web retrieval** and **chat history** for accurate, context-aware responses
- Developed an **adaptive chunking algorithm** to segment text into large and small chunks with context overlap, allowing more targeted retrieval without overall context loss
- Combined **BM25** and **vector-search** results with **RRF** for **RAG retrieval** in a **small-to-big** pipeline to improve search accuracy and context retention
- Built a **concurrent Python scraping engine** using **thread pools** to extract real-time web search payloads with **low latency** for live RAG augmentation

PROJECTS

🔗 **ASL Translator** | *Python, PyTorch, OpenCV, MediaPipe, CUDA*

April 2026 - Present

- Designed a **Landmark-to-Sequence** architecture isolating skeletal features from raw video, **reducing input dimensionality by >99%** to enable **real-time edge execution**
- Engineered a **Stacked LSTM network** in **PyTorch** for spatial-temporal modeling, prioritizing a **minimal VRAM footprint** over Transformer architectures to ensure real-time execution on local hardware
- Integrating **CTC loss** alongside a fine-tuned local **T5-Small Transformer** to decode unsegmented gestures and perform offline, real-time ASL-to-English syntax correction

Extracurriculars: UW Computer Science Club, UW Data Science Club, UW Game Dev Club, UWCSA

Hobbies/Interests: Piano, Violin, Guitar, Badminton, Swimming, Video Game Design, Computer Hardware